PROBLEM SET 7

1. Convert the following to minterms
   ABC + AB + DC + $\overline{D}$
   ABCDE + AB$\overline{E}$ + ACD

2. Convert the following to maxterms
   a. (A+B+$\overline{D}$) ($\overline{A}$+C+D) ($\overline{A}$+ $\overline{D}$)
   b. A ($\overline{A}$+B) ($\overline{C}$)

3. Reduce the following expressions using K-map
   ($\overline{A}$+B) (A+B+$\overline{D}$) (B+$\overline{C}$) (B+C+D)

4. Reduce ∏M(1, 2, 3, 5, 6, 7, 8, 9, 12, 13) and implement it in universal logic.

5. Reduce the following expressions using K-map and implement them in universal logic.
   (a) ∏M (3, 6, 8, 11, 13, 14) .d(1, 5, 7, 10)
   (b) $\boldsymbol{\Sigma}$m (0, 1, 4, 5, 6, 7, 9, 11, 15) + d (10, 14)

6. Simplify the following logic expressions and realize them using universal gates.
   a. $\boldsymbol{\Sigma}$m (0, 2, 3, 10, 12, 16, 17, 18, 21, 26, 27) + d (11, 13, 19, 20)
   b. $\boldsymbol{\Sigma}$m (4, 6, 8, 9, 10, 12, 13, 18, 19, 25, 26, 29, 33, 35, 36, 41, 42, 48, 49, 50, 56, 57) + d(0, 1, 11, 15, 30, 38, 40)

7. Minimize and implement the following multiple output functions in SOP form using K-maps.
   $f_1 = \Sigma$ m (2, 3, 7, 10, 11, 14) + d (1, 5, 15)
   $f_2 = \Sigma$ m (0, 1, 4, 7, 13, 14) + d (5,8,15)

   Hint: Use shared minterm and map to K-map

8. Reduce the following expressions using a 4-variable K-map.
   a. S = m_1+ F m_2+ G m_4+ m_6+ $\overline{F}$ m_7+ H m_9 + (F + G) m_{10} +$d[(F+\bar{G})m_0 + Gm_5 + m_{14} + \bar{F}m_{15}]$
   b. T = E m_0+ H m_1+ F m_2+ ($\overline{E}$ + G) m_6 + m_8 + d (m_{10} + $\overline{E}$ m_{14} + G m_7)
   c. U = E m_3 + m_5 + m_7 + F m_{10} + m_{12} + m_{14}